D20A **Roll No.19**

Experiment No. 8

Aim:

To configure a cryptocurrency wallet using MetaMask and integrate it with the ChronoVault DApp for wallet connection and blockchain interaction.

Theory:

A cryptocurrency wallet serves as a tool for managing blockchain accounts and authorizing transactions. MetaMask is a browser-based wallet that connects decentralized applications (DApps) to Ethereum-compatible networks and makes the selected account available to the application. It allows DApps to request account access and initiate transaction requests, while users maintain control over transaction approval through the wallet.

DApp integration typically requires connecting the wallet, choosing a supported blockchain network, and granting the necessary permissions. In this experiment, the ChronoVault DApp integrates MetaMask to connect a wallet and offers operations such as TOTP configuration, wallet deployment, funding, and transaction execution. The screenshots illustrate the wallet connection prompt, network authorization, connected-wallet interface, account information, and a smart contract deployment request on the Sepolia test network.

Steps:

1. Open the ChronoVault DApp at https://chronovault-ethonline.vercel.app.
2. Select MetaMask from the available wallet connection options.
3. Review the MetaMask connection request for the ChronoVault website and inspect the requested account permissions.
4. Continue to the permission review and verify that the requested network is Sepolia.
5. Confirm the wallet connection request in MetaMask.
6. Verify that the ChronoVault dashboard displays the connected wallet address and Connected status.
7. Open the account configuration and enter the ChronoVault account details shown in the setup interface.
8. Open the wallet deployment workflow and review the contract deployment request and transaction details presented by MetaMask.

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Output:

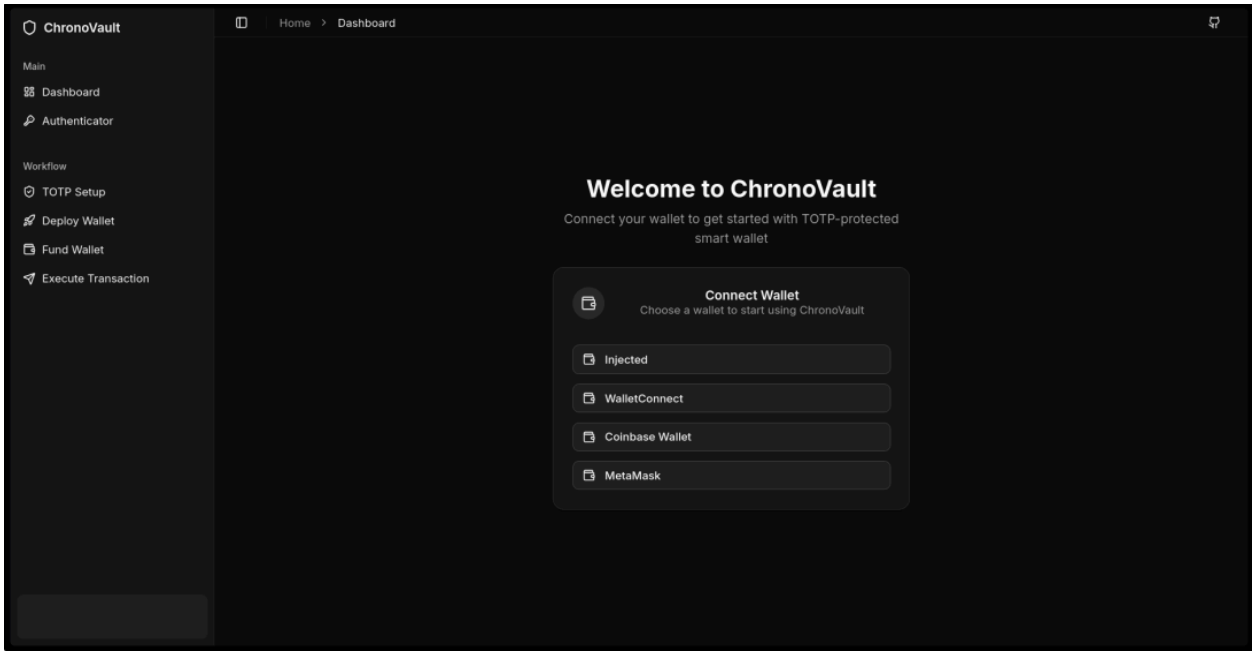


Fig. 1. ChronoVault wallet connection screen

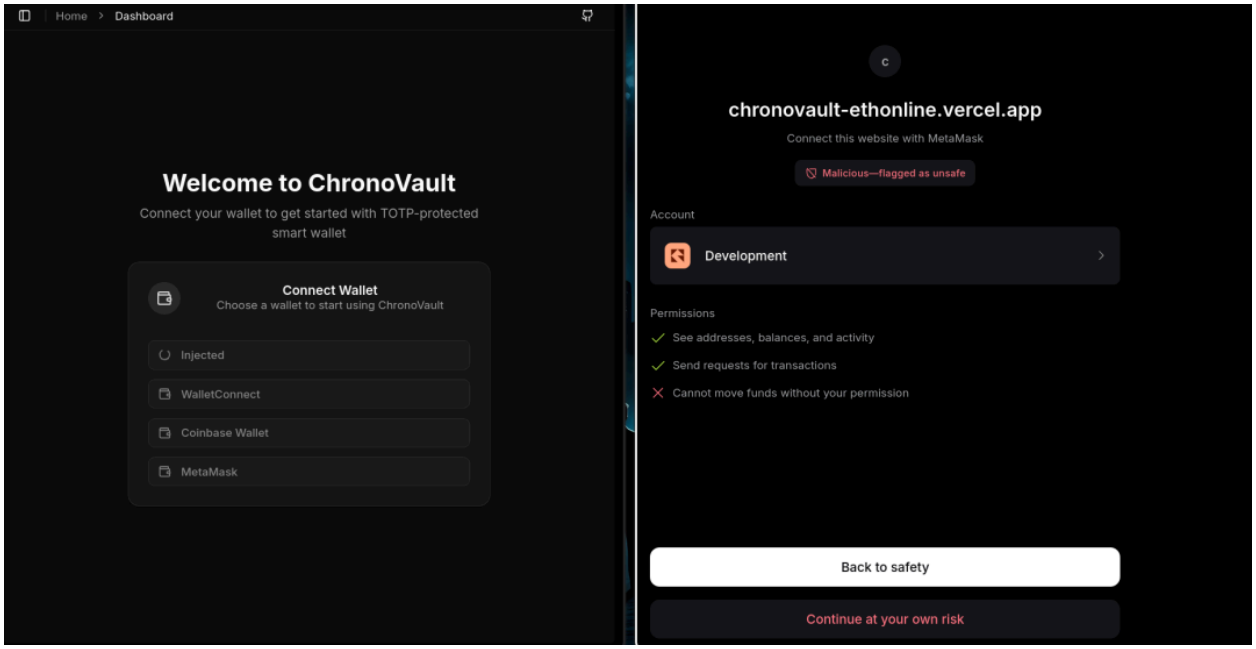


Fig. 2. MetaMask connection request for ChronoVault

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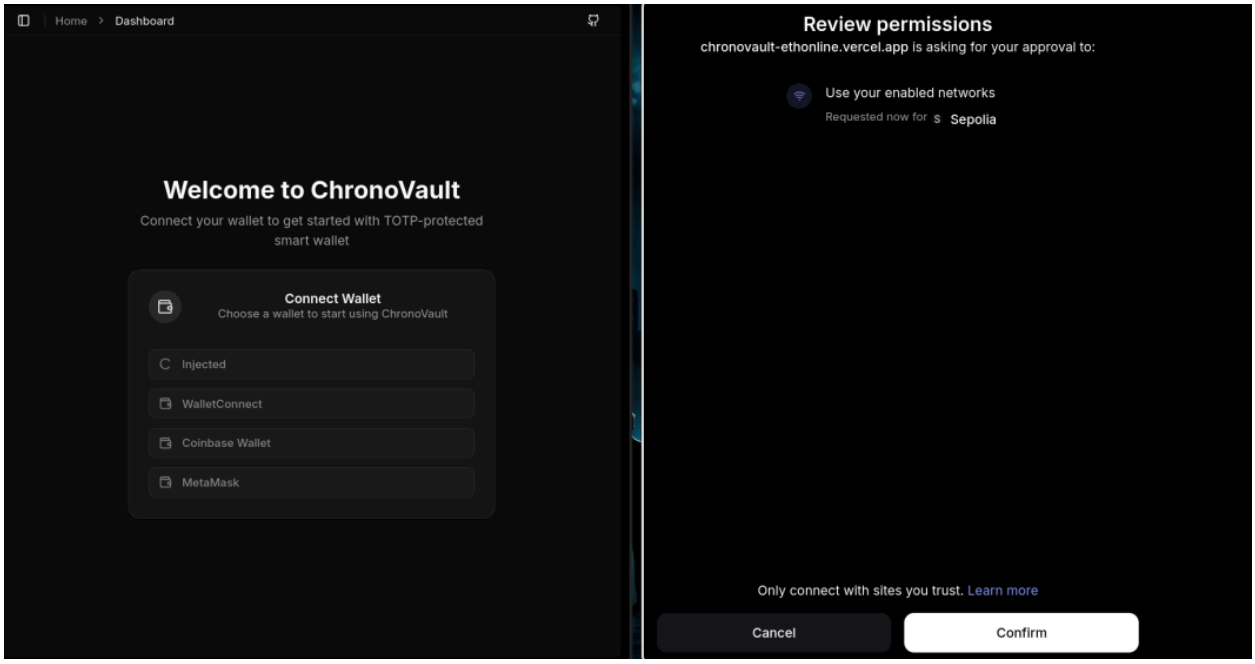


Fig. 3. MetaMask network permission confirmation

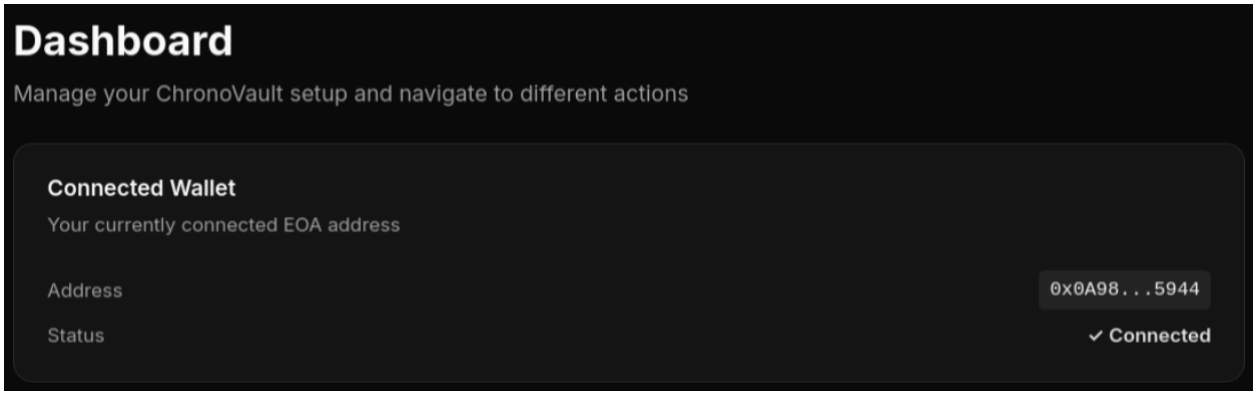


Fig. 4. Dashboard showing the connected wallet

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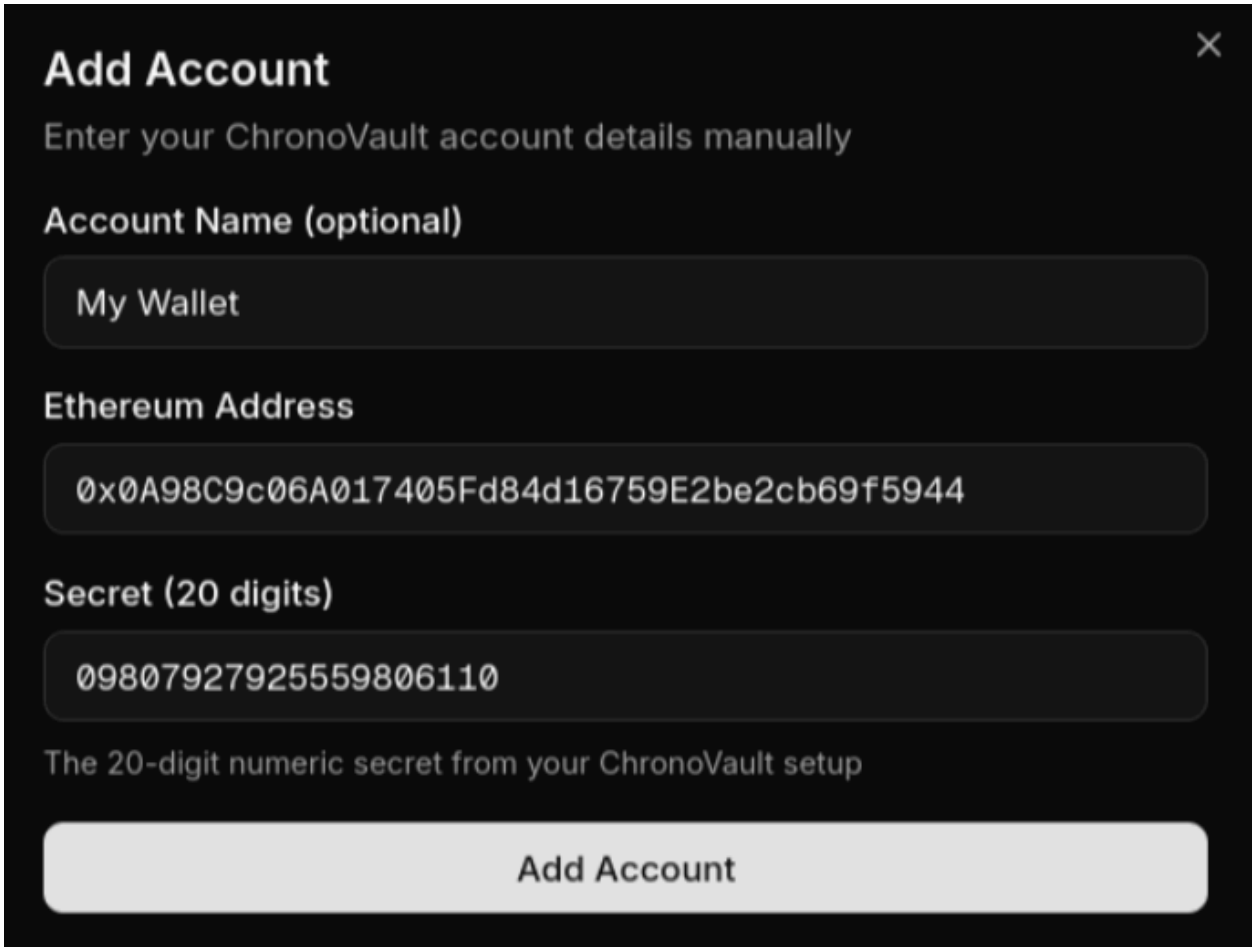


Fig. 5. ChronoVault account details form

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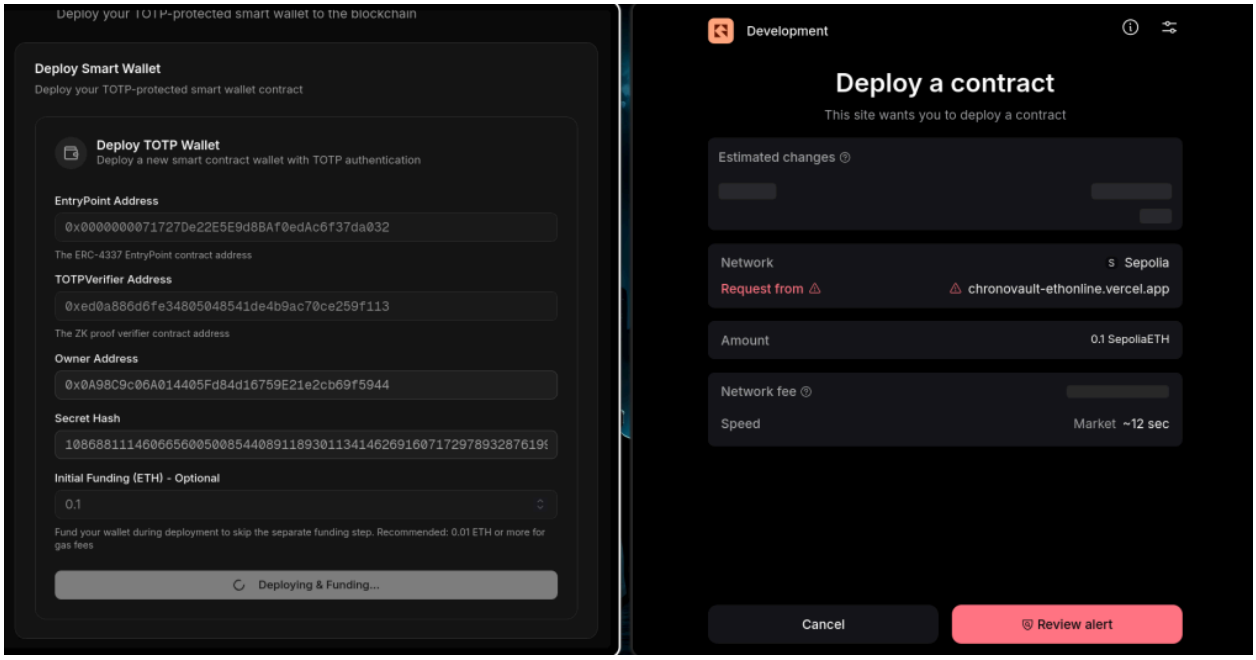


Fig. 6. Smart wallet deployment request in MetaMask

Conclusion:

The MetaMask wallet was successfully linked to the ChronoVault DApp, and the requested permissions for the Sepolia network were verified. The dashboard confirmed the wallet connection, and the account settings interface was accessed without issues. Additionally, the deployment process initiated a smart-wallet deployment request through MetaMask. This experiment demonstrated the successful integration of a browser-based cryptocurrency wallet with a decentralized application.